



THE KING'S SCHOOL

2008 Higher School Certificate Trial Examination

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 84

- Attempt Questions 1-7
- All questions are of equal value

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Total marks – 84

Attempt Questions 1-7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Find the remainder when the polynomial $P(x) = x^5 - 2x^3 + 11$ is divided by $x + 1$ **2**

(b) The interval AB has end points $A(-1, 8)$ and $B\left(\frac{10}{3}, \frac{28}{3}\right)$

Find the point $P(x, y)$ which divides AB externally in the ratio 3:2. **2**

(c) Evaluate $\int_0^{\frac{1}{4}} \frac{12}{\sqrt{1 - 4x^2}} dx$ **3**

(d) Find in simplest form the derivative of $\frac{x}{1 + x^2} + \tan^{-1} x$ **3**

(e) By writing $2x$ as $(x + y) + (x - y)$, or otherwise, simplify $\frac{2x}{x^2 - y^2} - \frac{1}{x - y}$ **2**

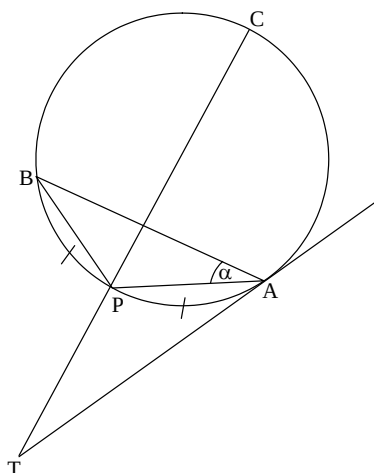
End of Question 1

-
- (a) (i) Find the remainder when $3x^4 + 2x^2 + 2x - 1$ is divided by $x^2 + 1$ **2**
- (ii) Hence find $\int \frac{3x^4 + 2x^2 + 2x - 1}{x^2 + 1} dx$ **1**
- (b) Use the substitution $u = 1 - x$ to evaluate $\int_0^1 360x(1 - x)^4 dx$ **3**
- (c) A particle moves on the x axis with velocity v given by $v = (x + 1)^2$. Initially the particle is at the origin. Find the initial acceleration. **3**
- (d) Solve the inequality $\frac{3x - 1}{2x + 3} > 1$ **3**

End of Question 2

- (a) Solve $12\sin^{-1}x = \cos^{-1}x$, giving your answer correct to two decimal places. **3**

(b)



TA is a tangent at A to the circle. TPC is a secant to the circle and B is chosen on the circle so that arc PA = arc PB. Let $\angle PAB = \alpha$

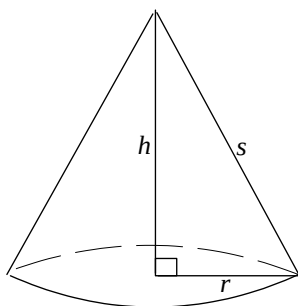
- (i) Explain why $\angle PBA = \alpha$ **1**
- (ii) Prove that AP bisects $\angle TAB$ **2**
- (c) A particle is moving in simple harmonic motion on the x axis according to the equation of motion $x = -12 \cos nt$, where $t \geq 0$ is time and $n > 0$. The period of the motion is T .
- (i) Prove that the particle first reaches the position $x = 6$ when the time is $\frac{T}{3}$ **3**
- (ii) Find the velocity of the particle when $t = \frac{T}{3}$ **2**
- (iii) Write down the time that will elapse after $t = \frac{T}{3}$ for the particle to be next at rest. **1**

End of Question 3

- (a) A particle is falling vertically so that at any time t s its velocity v m/s is given by $\frac{dv}{dt} = 10 - 0.2v$. Initially $v = 20$.

- (i) Verify by differentiation that $v = 50 + Ae^{-0.2t}$ for some constant A . 2
- (ii) Find A . 1
- (iii) Find the time taken, correct to one decimal place, for the velocity to reach 40 m/s. 2
- (iv) Find the distance travelled during the first 5 seconds. 2

(b)



The diagram shows a cone with base radius r , height h and slant height s .

A conical pile is being formed at a constant rate of $2\text{m}^3/\text{min}$.

The pile at any time t is such that $h = \frac{4}{3}r$.

- (i) Show that $s = \frac{5r}{3}$ 1
- (ii) Find the rate of increase of the curved surface area at the instant the radius is 5m.

$$[V = \frac{1}{3}\pi r^2 h, \text{ Curved Surface Area} = \pi r s] \quad 4$$

End of Question 4

-
- (a) Find the coefficient of x^9 in the binomial expansion of $\left(2x + \frac{3}{x}\right)^{30}$
[LEAVE YOUR ANSWER IN UNSIMPLIFIED FORM] **4**
- (b) (i) Sketch $y = \sin x$ and $y = \cos x$ on the same axes for $0 \leq x \leq \frac{\pi}{2}$, clearly showing their point of intersection. **2**
- (ii) The region enclosed between $y = \sin x$ and $y = \cos x$ and the vertical lines $x = 0$ and $x = \frac{\pi}{2}$ is revolved about the x axis.

Find the volume of the solid generated. **3**
- (c) Prove by induction for integers $n \geq 1$ that
 $4.1 + 8.3 + 12.3^2 + \dots + 4n.3^{n-1} = (2n-1)3^n + 1$ **3**

End of Question 5

(a) Let $f(t) = t^3 - 12t - 2$.

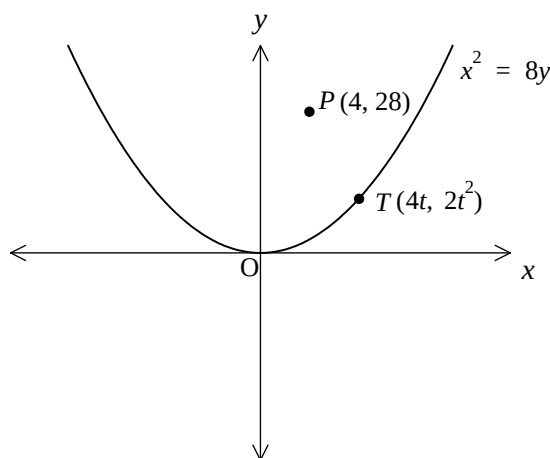
Since $f(-1) = 9$ and $f(0) = -2$ there is a root of $f(t) = 0$ between $t = -1$ and $t = 0$.

[DO NOT EXPLAIN THE REASON]

Use Newton's Method once with a trial root of $t = -0.2$ to give a two decimal approximation to the root between $t = -1$ and $t = 0$.

3

(b)



The diagram shows the parabola $x^2 = 8y$ and the point $P(4, 28)$ interior to the parabola. Let $T(4t, 2t^2)$ be any point on the parabola.

- (i) Prove that the equation of the normal at T is $x + ty = 4t + 2t^3$ **2**
- (ii) The normal at T passes through $P(4, 28)$. Show that $t^3 - 12t - 2 = 0$ **1**
- (iii) Deduce that if there are three normals which can pass through $P(4, 28)$ then the sum of the x coordinates of the points $T(4t, 2t^2)$ is zero. **2**
- (iv) Find the sum of the y coordinates of the points $T(4t, 2t^2)$ in (iii). **2**
- (v) Prove that there are three normals which pass through $P(4, 28)$. **2**

End of Question 6

(a) Let $f(x) = e^x - e^{-x}$

(i) Show that the function is an odd function . **1**

(ii) Show that $f(x)$ increases as x increases for all values of x . **1**

(iii) Sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same diagram and include the line $y = x$. **2**

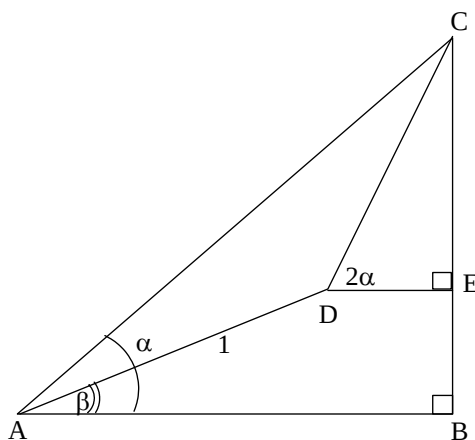
(iv) Show that $e^{\ln x} = x$ **1**

(v) It can be shown that $f^{-1}(x) = \ln \left(\frac{x + \sqrt{x^2 + 4}}{2} \right)$

[DO NOT SHOW THIS]

Show that $\int_0^2 f^{-1}(x) dx = 2\ln(1 + \sqrt{2}) + 2 - 2\sqrt{2}$ **3**

(b)



In the diagram

$AD = 1$, $\angle BAC = \alpha$, $\angle BAD = \beta$ and $\angle EDC = 2\alpha$,
 $\angle CED = \angle CBA = 90^\circ$

(i) Find $\angle ACD$. **1**

(ii) Prove that $BC = \sin(2\alpha - \beta)$. **3**

End of Examination

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Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note: $\ln x = \log_e x, \quad x > 0$



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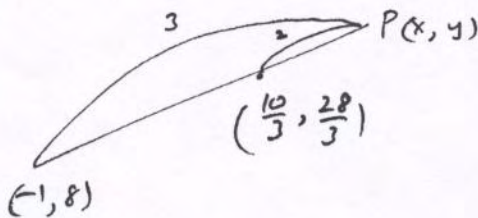
2008 Higher School Certificate Trial Examination

Mathematics Extension 1

Question	Algebra and Number	Geometry	Functions	Trigonometry	Differential Calculus	Integral Calculus	Total
1	(e) 2		(a), (b) 4		(d) 3	(c) 3	12
2	(d) 3		(a)(i) 2		(c) 3	(a)(ii), (b) 4	12
3		(b) 3	(a) 3	(c) 6			12
4			(a)(ii)(iii) 3	(a)(i) 2	(b) 5	(a)(iv) 2	12
5	(a), (c) 7			(b)(i) 2		(b)(ii) 3	12
6			(a), (b) 12				12
7		(b)(i) 1	(a) 8	(b)(ii) 3			12
Total	12	4	32	13	11	12	84

Question 1

$$(a) R = P(-1) = -1 + 2 + 11 = 12$$

(b)  $\therefore x = \frac{3 \cdot \frac{10}{3} - 2(-1)}{3 - 2} = 10 + 2 = 12$

$$y = 28 - 16 = 12$$

$$\therefore P = (12, 12)$$

$$(c) I = \frac{12}{2} [\sin^{-1} 2x]_0^{\frac{1}{4}} = 6 \left(\frac{\pi}{6} - 0 \right) = \pi$$

$$(d) \frac{1+x^2 - x(2x)}{(1+x^2)^2} + \frac{1}{1+x^2}$$

$$= \frac{1-x^2}{(1+x^2)^2} + \frac{1+x^2}{(1+x^2)^2} = \frac{2}{(1+x^2)^2}$$

$$(e) \frac{x+y + x-y}{(x-y)(x+y)} - \frac{1}{x-y} = \frac{1}{x-y} + \frac{1}{x+y} - \frac{1}{x-y} = \frac{1}{x+y}$$

or $\frac{2x}{(x-y)(x+y)} - \frac{x+y}{(x-y)(x+y)}$

$$= \frac{x-y}{(x-y)(x+y)} = \frac{1}{x+y}$$

Question 2

(a) (i) $x^2 + 1 \overline{) \begin{array}{r} 3x^2 - 1 \\ 3x^2 \\ \hline -2x - 1 \\ -2x \\ \hline 0 \end{array}}$ $\therefore \text{remainder} = 2x$

(ii) $I = \int \frac{3x^2 - 1}{x^2 + 1} dx = \int \left(3 - \frac{1}{x^2 + 1} \right) dx = 3x - \ln(x^2 + 1) + C$

(b) $u = 1 - x$; $x = 0, u = 1$
 $\frac{du}{dx} = -1$; $x = 1, u = 0$

$\therefore I = - \int_1^0 360 (1-u) u^4 du$
 $= 360 \int_0^1 u^4 - u^5 du = 360 \left[\frac{u^5}{5} - \frac{u^6}{6} \right]_0^1$
 $= 360 \left(\frac{1}{5} - \frac{1}{6} \right) = 360 \cdot \frac{1}{30} = 12$

(c) $\frac{1}{2} v^2 = \frac{1}{2} (x+1)^4$

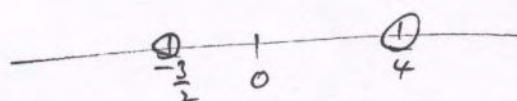
$\therefore \ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 2(x+1)^3$

$\therefore$ initial acceln at $x=0$ is 2

(d) (LOTS OF ALTERNATIVES) MOST COMMON ...

$2x+3 \neq 0$ i.e. $x \neq -\frac{3}{2}$

Put $\frac{3x-1}{2x+3} = 1 \Rightarrow 3x-1 = 2x+3$
 $x = 4$

$\Rightarrow$ 

Try $x=0 \Rightarrow -\frac{1}{3} > 1$ X

$\therefore x < -\frac{3}{2}$ or $x > 4$

Question 3

$$(a) \therefore 12 \sin^{-1} x = \frac{\pi}{2} - \sin^{-1} x$$

$$13 \sin^{-1} x = \frac{\pi}{2} \Rightarrow \sin^{-1} x = \frac{\pi}{26}$$

$$\text{or } x = \sin \frac{\pi}{26} = 0.12, 2 \text{ d.p.}$$

(b) (i) $\angle PBA = \angle PAB = d$, angles at circumference standing on same size arc, $PA = PB$

(ii) $\angle TAP = \angle PBA = d$, alternate segment thm

But $\angle PAB = d$, data

$\therefore AP$ bisects $\angle TAB$

$$(c) (i) T = \frac{2\pi}{n} \Rightarrow n = \frac{2\pi}{T}$$

$$\therefore 6 = -12 \cos \frac{2\pi}{T} t \Rightarrow \cos \frac{2\pi}{T} t = -\frac{1}{2}$$

$$\therefore \frac{2\pi}{T} t = \frac{2\pi}{3} \text{ for first } t$$

$$\Rightarrow t = \frac{T}{3}$$

$$(ii) v = -12 (-n \sin nt) = \frac{24\pi}{T} \sin \frac{2\pi}{3}$$

$$= \frac{24\pi}{T} \cdot \frac{\sqrt{3}}{2} = \frac{12\sqrt{3}\pi}{T}$$

(iii) since period = T the time that will elapse

$$= \frac{T}{2} - \frac{T}{3} = \frac{T}{6}$$

Question 4

(a) (i) If $v = 50 + A e^{-0.2t}$ then

$$\begin{aligned}\frac{dv}{dt} &= -0.2 A e^{-0.2t} = -0.2(v-50) \\ &= 10 - 0.2v\end{aligned}$$

(ii) $t=0, v=20 \Rightarrow 20 = 50 + A, A = -30$

(iii) $\therefore 40 = 50 - 30 e^{-0.2t}$

$$e^{-0.2t} = \frac{1}{3}$$

$$\text{or } e^{0.2t} = 3$$

$$\therefore 0.2t = \ln 3 \Rightarrow t = 5.5 \text{ s, 1 d.p.}$$

(iv) $x = \int_0^5 50 - 30 e^{-0.2t} dt$

$$= [50t + 150 e^{-0.2t}]_0^5$$

$$= 250 + 150 e^{-1} - 150$$

$$= 100 + 150 e^{-1} \text{ m}$$

$$[= 155 \text{ m, nearest metre}]$$

(b) (i) $s^2 = h^2 + r^2 = \frac{16r^2}{9} + r^2 = \frac{25r^2}{9} \Rightarrow s = \frac{5r}{3}$

(iii) $A = \pi r s = \pi r \cdot \frac{5r}{3} = \frac{5}{3} \pi r^2 \quad \therefore \frac{dA}{dr} = \frac{10}{3} \pi r$

$$V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi r^2 \cdot \frac{4}{3} r = \frac{4}{9} \pi r^3 \quad \therefore \frac{dV}{dr} = \frac{4}{3} \pi r^2$$

$$\therefore \frac{dA}{dt} = \frac{dA}{dr} \cdot \frac{dr}{dt} = \frac{dA}{dr} \cdot \frac{dr}{dV} \cdot \frac{dV}{dt} \quad \text{where } \frac{dV}{dt} = 2$$

$$= \frac{10}{3} \pi r \cdot \frac{3}{4\pi r^2} \cdot 2 = \frac{5}{r} = 1 \text{ m}^2/\text{min} \text{ when } r=5$$

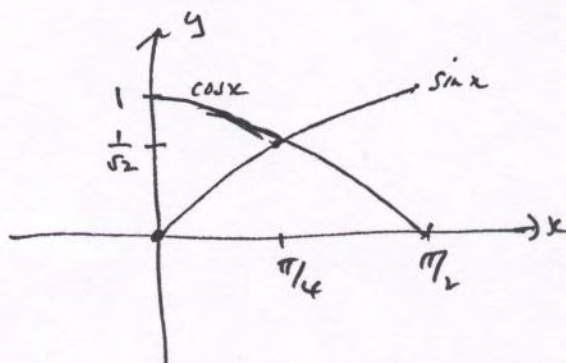
Question 5

$$\begin{aligned}(a) \quad u_{k+1} &= \binom{30}{k} (2x)^{30-k} \left(\frac{3}{x^2}\right)^k \\&= \binom{30}{k} 2^{30-k} 3^k \frac{x^{30-k}}{x^{2k}} \\&= \binom{30}{k} 2^{30-k} 3^k x^{30-3k}\end{aligned}$$

For coefficient of x^9 we'd have $30-3k=9$
 $\Rightarrow k=7$

$\therefore$ coefft of x^9 is $\binom{30}{7} 2^{23} 3^7$

$$(b) (i) \sin x = \cos x \Rightarrow \tan x = 1 \Rightarrow x = \frac{\pi}{4} \text{ for } 0 < x < \frac{\pi}{2}$$



$$\begin{aligned}(ii) \text{ From sketch, } V &= 2\pi \int_0^{\pi/4} \cos^2 x - \sin^2 x \, dx \\&= 2\pi \int_0^{\pi/4} \cos 2x \, dx \\&= 2\pi \left[\frac{\sin 2x}{2} \right]_0^{\pi/4} \\&= \pi (1-0) = \pi\end{aligned}$$

(C) For $n=1$, $LS = 4 \times 1 = 4$

$$RS = 1 \times 3 + 1 = 4$$

$\therefore$ Assume $4 \cdot 1 + 8 \cdot 3 + \dots + 4n \cdot 3^{n-1} = (2n-1)3^n + 1$ for any integer $n \geq 1$

Then $4 \cdot 1 + 8 \cdot 3 + \dots + 4n \cdot 3^{n-1} + 4(n+1)3^n$

$$= (2n-1)3^n + 1 + (4n+4)3^n, \text{ using the assumption}$$

$$= 3^n(2n-1 + 4n+4) + 1$$

$$= 3^n(6n+3) + 1$$

$$= 3^{n+1}(2n+1) + 1 = RS \text{ for } n+1$$

$\therefore$ if correct for n , it's correct for $n+1$

But it is correct for $n=1$

$\therefore$ by induction, $4 \cdot 1 + \dots + 4n \cdot 3^{n-1} = (2n-1)3^n + 1$ for $n \geq 1$

Question 6

(a) $f'(t) = 3t^2 - 12$

$$\therefore t_1 = -0.2 - \frac{(-0.2)^3 - 12(-0.2) - 2}{3(-0.2)^2 - 12} = -0.17, \text{ 2 d.p.}$$

(b) (i) $y = \frac{x^2}{8} \therefore \frac{dy}{dx} = \frac{2x}{8} = \frac{x}{4} = t \text{ at } T(4t, 2t^2)$

$\therefore$ normal is $y - 2t^2 = -\frac{1}{t}(x - 4t)$

or $ty - 2t^3 = -x + 4t$

i.e. $x + ty = 4t + 2t^3$

(ii) Since $P(4, 28)$ is on the normal

$$4 + 28t = 4t + 2t^3$$

i.e. $t^3 - 12t - 2 = 0$

(iii) Sum of x coordinates $= 4(t_1 + t_2 + t_3)$, say,

where $t_1 + t_2 + t_3 = 0$, sum of roots of equation in (ii)

~~sum~~ sum of x coordinates $= 0$

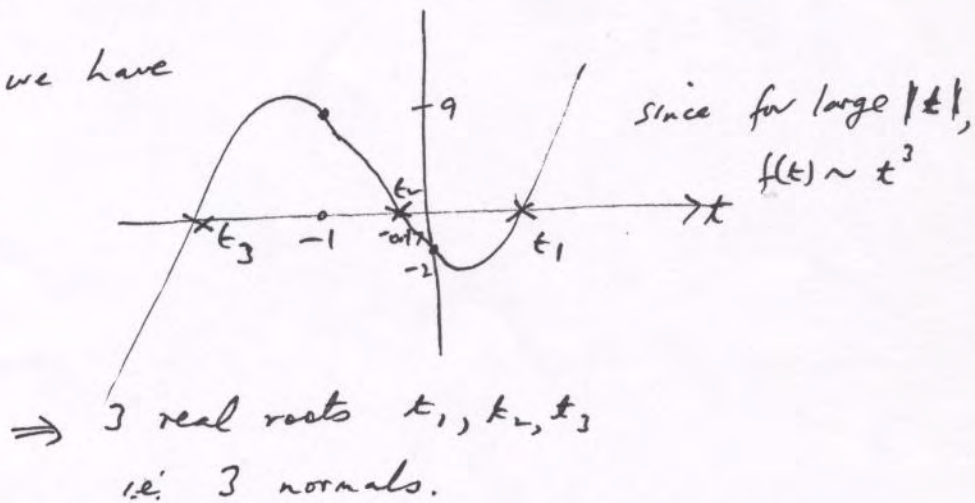
(iv) Sum $= 2(t_1^2 + t_2^2 + t_3^2)$

$$= 2((t_1 + t_2 + t_3)^2 - 2(t_1 t_2 + t_2 t_3 + t_3 t_1))$$

$$= 2(0^2 - 2(-12)) = 48$$

(v) Need to show $t^3 - 12t - 2 = 0$ has 3 real roots

From (a) we have



Question 7

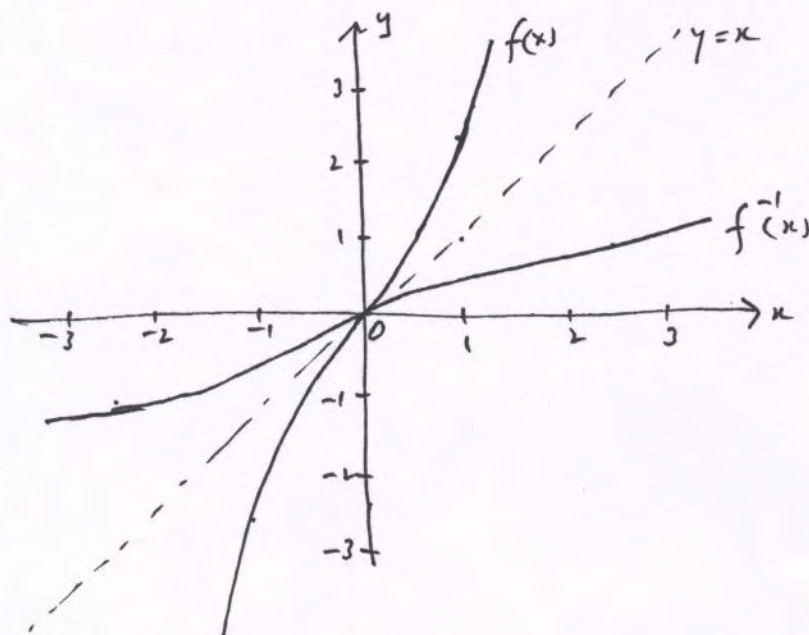
(a) (i) $f(-x) = e^{-x} - e^x = -(e^x - e^{-x}) = -f(x)$

$\therefore f(x)$ is an odd function

(ii) $f'(x) = e^x + e^{-x} > 0 \quad \forall x$

$\therefore f(x)$ is an increasing function $\forall x$

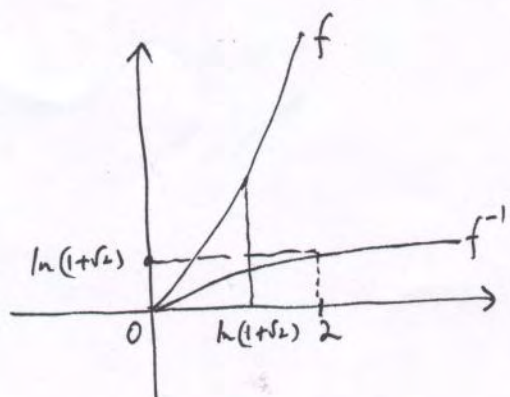
(iii) $f(0) = 0, f(1) = e - e^{-1} \approx 2.35, f'(0) = 2$



(iv) Suppose $e^{\ln x} = N$

Then $\ln x = \log_e N \Rightarrow N = x$ i.e. result

(v) $f^{-1}(2) = \ln\left(\frac{2+\sqrt{8}}{2}\right) = \ln(1+\sqrt{2})$



$$\therefore \int_0^2 f^{-1}(x) dx = 2 \ln(1+\sqrt{2}) - \int_0^{\ln(1+\sqrt{2})} e^x - e^{-x} dx$$

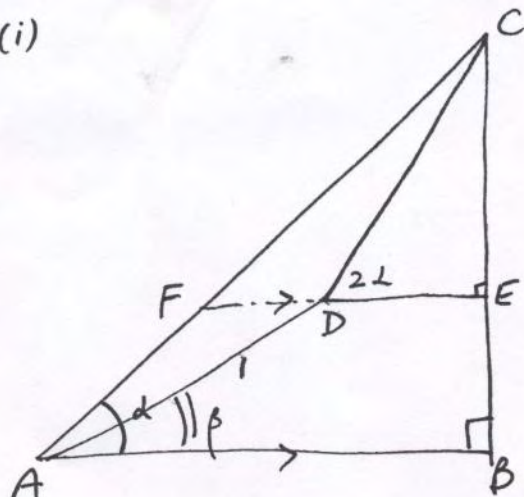
$$= 2 \ln(1+\sqrt{2}) - [e^x + e^{-x}]_0^{\ln(1+\sqrt{2})}$$

$$= 2 \ln(1+\sqrt{2}) - \left(1+\sqrt{2} + \frac{1}{1+\sqrt{2}} - (1+1)\right)$$

$$= 2 \ln(1+\sqrt{2}) - (1+\sqrt{2} + \sqrt{2}-1-2)$$

$$= 2 \ln(1+\sqrt{2}) + 2 - 2\sqrt{2}$$

(b) (i)



Produce ED to meet AC at F

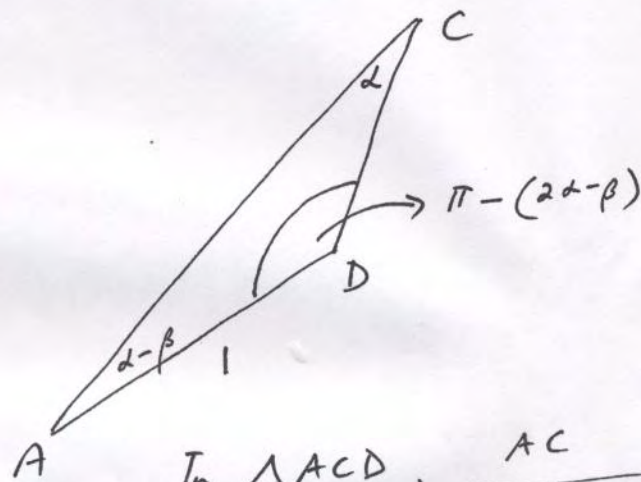
$$\text{Then } \angle CFE = \alpha (= \angle CAB)$$

$\therefore \angle ACD = \alpha$, ext $\angle$ thm in $\triangle FDC$

[No reasons needed anywhere]

[MANY ALTERNATIVES]

(ii) In $\triangle CAB$, $\sin \alpha = \frac{BC}{AC}$ ie. $BC = AC \sin \alpha$



$$\text{In } \triangle ACD, \frac{AC}{\sin(\pi - (2\alpha - \beta))} = \frac{1}{\sin \alpha}$$

$$\Rightarrow AC = \frac{\sin(2\alpha - \beta)}{\sin \alpha}$$

$$\therefore BC = \frac{\sin(2\alpha - \beta)}{\sin \alpha} \cdot \sin \alpha = \sin(2\alpha - \beta)$$